

Credit Card Fraud Detection Using Predictive Data Mining Techniques

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1. Introduction

Credit card fraud is a serious challenge for financial institutions, with fraudulent transactions hidden among a vast number of legitimate ones. Effective fraud detection requires sophisticated data-mining models capable of identifying extremely rare fraud patterns while avoiding false alarms. In this coursework, our team aims to develop and evaluate a complete predictive modelling system for fraud detection. This includes data preprocessing, imbalance handling, model comparison, and rigorous validation protocols necessary for high-risk domains such as finance. Our goal is to construct an end-to-end system that not only builds predictive models but also addresses issues commonly reported in fraud-detection research, such as extreme class imbalance, PCA-anonymised features, and evaluation challenges. The final system will generate benchmark results, compare multiple modelling techniques, and document limitations and implications for real-world deployment.

2. Brief

What are we going to do?

We plan to build an **end-to-end predictive modelling system** for detecting fraudulent credit card transactions using the *Credit Card Fraud Detection* dataset provided by the Machine Learning Group at ULB. [1]

The system will include:

- Data ingestion and profiling
- Handling extreme imbalance (0.172% fraud cases)
- Preprocessing of PCA-transformed numerical features
- Multiple modelling approaches (supervised + anomaly detection)
- Cross-validation and robust evaluation
- Interpretability and performance analysis
- Discussion of limitations and risks

What datasets will we use?

We will use the **Credit Card Fraud Detection Dataset** from Kaggle, originally published by the Machine Learning Group at Université Libre de Bruxelles (ULB). The dataset consists of **284,807 transactions**, covering a two-day period in September 2013, with **492 fraudulent cases** (0.172%). [1]

Features include:

- **Time** and **Amount** (original numeric variables)
- **V1–V28**, which are PCA-transformed components
anonymization of confidential fields
- **Class**, indicating fraudulent (1) or legitimate (0) transactions [1]

What approaches will we try?

Our modelling approach will begin with supervised algorithms such as logistic regression, random forests, and gradient-boosting models, which are commonly applied to fraud-detection problems. Because most features (V1–V28) are anonymised through **PCA transformation**, we anticipate limited interpretability but still expect meaningful performance comparisons across these models. [1]

Alongside supervised models, we intend to evaluate anomaly-based methods like **Isolation Forest** and **One-Class SVM**, as these techniques are well suited for detecting rare and unusual patterns in highly imbalanced datasets. To address the severe imbalance directly, we plan to experiment with undersampling, synthetic oversampling strategies such as **SMOTE**, and class-weight adjustments, allowing us to observe how each technique influences model performance.

What do we hope to find out?

- Identify which modelling techniques (supervised vs anomaly detection) perform best on highly imbalanced fraud-detection tasks.
- Understand the effect of different imbalance-handling strategies on model performance.
- Discover trade-offs between recall (catching fraud) and precision (avoiding false alarms).
- Evaluate limitations caused by PCA-transformed features (e.g., reduced interpretability).

This investigation will provide insights into developing effective fraud-detection models for financial applications and highlight challenges in deploying such systems.

3. References

- [1] Machine Learning Group, ULB. *Credit Card Fraud Detection Dataset* (European Cardholders, September 2013). Kaggle. <https://www.kaggle.com/datasets/mlg-ulb/creditcardfraud>